Bidding Away the Incentives to Drill

Revenue-Sharing Auctions in Mexico's Oil Industry

Tomoki Nishiyama

EPIC

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Revenue-Sharing Auction

- A government leases oil fields to private companies
- Bid: each company submits a revenue share r, the share of future revenue it pays to the government
  - Companies compete on a revenue tax rate
- The company with the highest rate r wins the contract
- The winner then chooses its drilling investment I given its committed rate
  - Winner's payoff $(1 - r)\, P\, F(I) - c(I)$
  - Government's payoff $r\, P\, F(I)$

RQ: How Does Auction Design Affect the Winner's Drilling Decision?

- Auction design choices affect the winning bid
- Through the winner's objective function, they also affect its investment and production
- Example: revenue-sharing with a minimum bid
  - The government raises the minimum bid to increase the winning bid r
  - This can discourage the winner's drilling investment I by reducing its return on investment
  - Lower investment leads to lower production $F(I)$
  - The government may lose revenue $r\,P\,F(I)$, since $r \uparrow$ but $I \downarrow$ and $F(I) \downarrow$

Why Should We Study Revenue-Sharing Auctions?

Common in resource auctions in emerging markets

- Oil in Mexico and Brazil, coal in India
- Used where other auction formats (FPA, profit-sharing auction) are not feasible

Same logic in pay-as-produced PPA auctions

- Power plants bid an electricity price
- The lowest price wins, and the plant then chooses how much to generate at that price
- A lower price can discourage generation

Implications for auction design

- The winning bid sets the winner's incentives after the auction
- Auction design should consider post-auction incentives, not just bidding

Contribution

Bhattacharya, Ordin, and Roberts (2022): pre-determined revenue shares

- Bonus auction with a pre-determined revenue share
- A higher revenue share delays drilling

Nguyen (2026): information disclosure in bonus auctions

- First-price bonus auction
- The winner learns about reserves from losing bids
- Disclosing losing bids encourages exploration

This paper: how revenue-sharing auction design affects the winner's drilling

Roadmap

Research Question

Data: Mexico's Oil Auctions

Reduced-Form Evidence: Effects of the Winning Bid on Drilling and Revenue

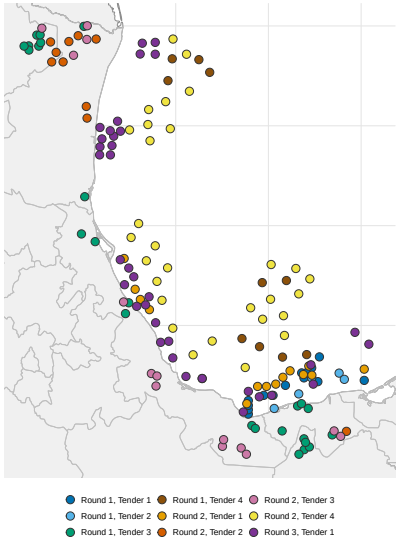
Structural Model: Bidding and Drilling

Counterfactuals: Alternative Auction Designs

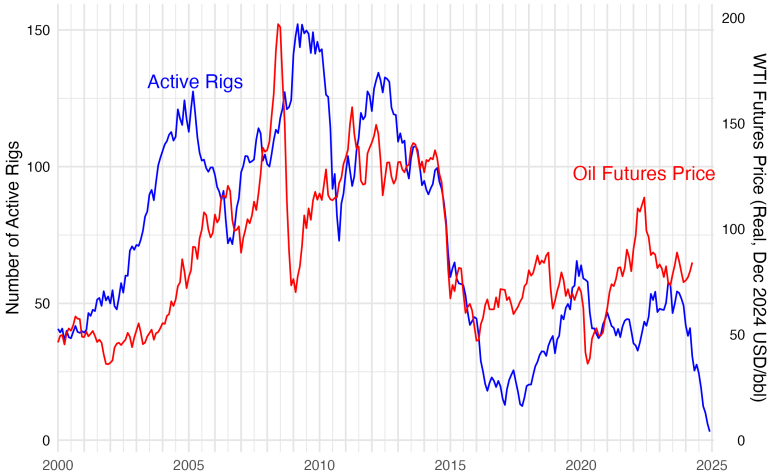
Conclusion

Revenue-Sharing Auctions in Mexico

- 157 blocks in 9 batches
- Firms bid a revenue share r, and the highest r wins
- r must lie between a floor and a cap (e.g. 3%–40%)
- Registered firms may stay out
- Exploration period of 5–10 years, then a production period of 20 years
- Data
  - All bids
  - Winner's monthly drilling investment and production
  - Monthly oil price
  - Government revenue



Drilling Tracks the Oil Price



Firms drill more when they expect the oil price to rise

Drilling and Production

Drilling

- Boring a hole through the Earth's crust to extract oil and gas
- Oil cannot be produced without drilling
- Drilling also reveals information about the field, which is useful for neighboring fields (Kellogg, 2011)

Production

- Production is constrained by drilling activity (Anderson, Kellogg, & Salant, 2018)
- Production costs almost nothing once the well is drilled

Awarded Licences by Stratum

Stratum	Awarded licences	Within 6 years	Within the exploration period
Onshore	21	12 (57.1%)	12 (57.1%)
Deep water	27	2 (7.4%)	3 (11.1%)
All	48	14 (29.2%)	15 (31.3%)

More than half of the licences end up with no drilling within the exploration period

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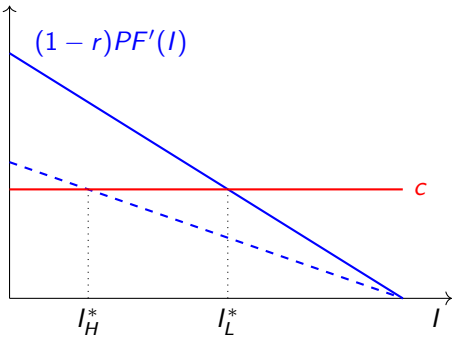
Revenue-Sharing and the Winner's Investment Incentives

Revenue-sharing ($100r$%)

$$\max_I \ (1-r)P\,F(I) - cI$$

$$(1-r)P\,F'(I^*) = c$$

A higher r lowers investment

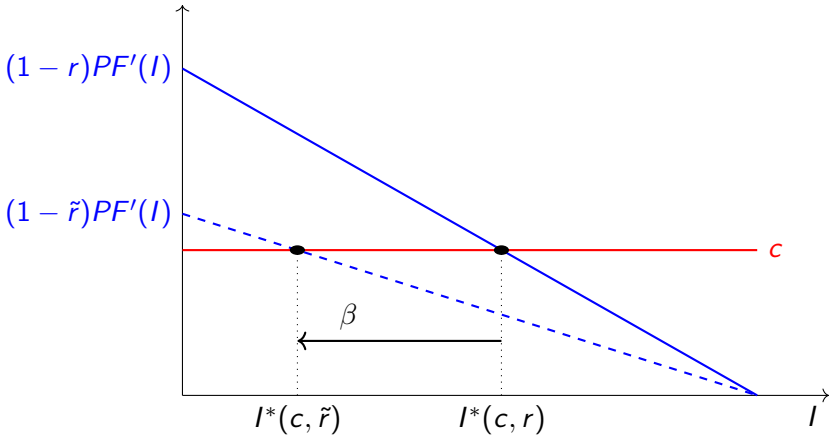


Solid lines: low r. Dashed lines: high r.

How can we connect this theory to the data?

$\text{Drilling}_\ell = \beta\,\text{Winning revenue share}_\ell + \varepsilon_\ell$?

What if we regress drilling investment I on the winning revenue share r by OLS?

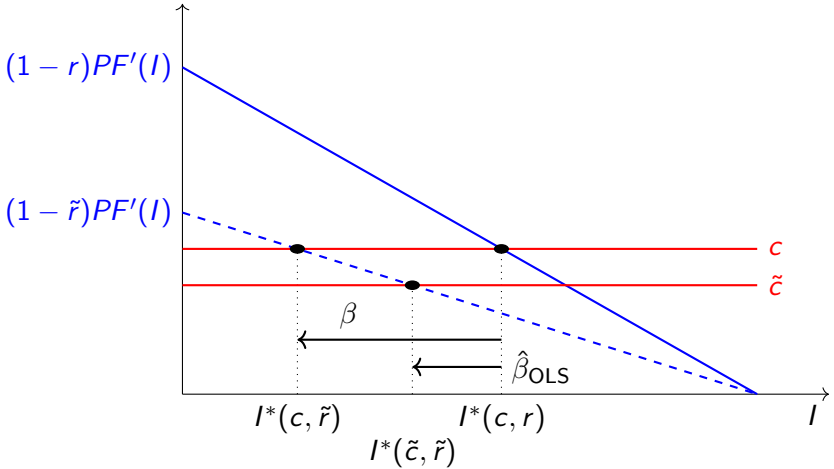


A firm that bids a higher r is likely to have a lower drilling cost c

$\text{Drilling}_\ell = \beta\,\text{Winning revenue share}_\ell + \varepsilon_\ell$

ε_ℓ contains the winner's cost C_ℓ, which is correlated with r_ℓ

$\hat{\beta}_{\text{OLS}}$ is biased toward zero $\Rightarrow$ need exogenous variation in the winning bid r_ℓ



IV for the Winning Bid

IV for the winning bid: the mean floor and cap of the other blocks in the same batch

$$F_{-\ell} = \frac{1}{n_t - 1} \sum_{j \in t, j \neq \ell} F_j \qquad C_{-\ell} = \frac{1}{n_t - 1} \sum_{j \in t, j \neq \ell} C_j$$

- High floors and caps on rival blocks push bidders toward block ℓ
  - First stage: $r_\ell = \underset{(2.59)}{11.94} + \underset{(0.21)}{0.89}\, F_{-\ell} + \underset{(0.06)}{0.31}\, C_{-\ell}$, F-stat : 61.9
- Assumption: block characteristics are independent within a batch
  - Floors and caps set for other blocks are uncorrelated with block ℓ's unobservable ε_ℓ

A Higher Winning Bid Delays Exploration but Raises Government Revenue

	Drilling				Production	Revenue
	(1)	(2)	(3)	(4)	(5)	(6)
	Years to drilling	log Expl. investment	Expl. plan-years	Wells drilled	Produced	Government
Unit	*years*	*US$m*	*count*	*count*	*binary*	*US$m*
Panel A. OLS						
Winning bid (%)	0.088***	0.005	-0.046***	0.094	0.008***	2.182*
	(0.016)	(0.006)	(0.013)	(0.063)	(0.002)	(1.180)
Panel B. 2SLS, winning bid instrumented by the rival-auction mean floor and mean cap						
Winning bid (%)	0.119***	0.002	-0.057***	0.059	0.005	4.508**
	(0.023)	(0.009)	(0.018)	(0.087)	(0.003)	(2.175)
First-stage F	61.9	61.9	61.9	61.9	61.9	61.9
Total obs.	104	76	104	104	104	93
Mean of dep. variable	3.913	4.127	4.510	6.308	0.365	28.093

$$r \uparrow \Rightarrow \; I^*(r) \downarrow \quad \text{and} \quad rPF(I^*(r)) \uparrow$$

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Common Oil Reserves and Heterogeneous Drilling Costs

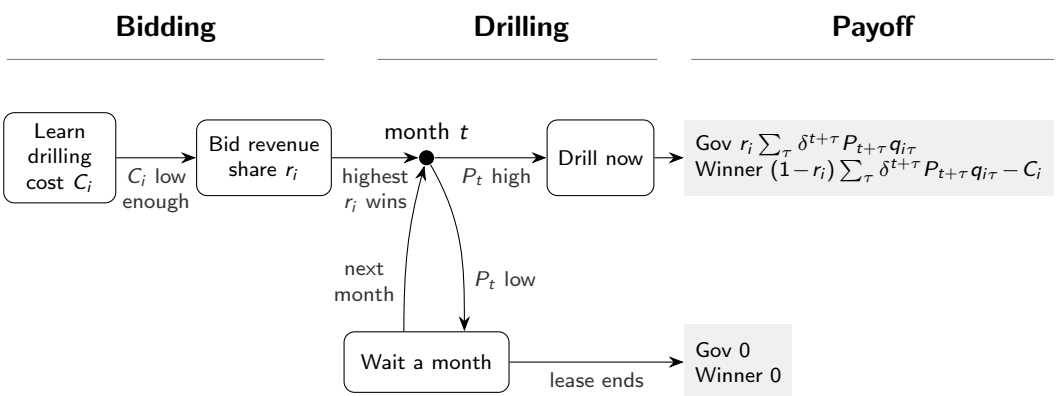
A government runs simultaneous revenue-sharing auctions, one per block ℓ.

- All bidders know the block's expected reserve quantity Q_ℓ (rational expectations)
- Bidders differ in their drilling cost $C_{i\ell}$, which is private information
  - Type $Q_\ell / C_{i\ell}$ (barrels per dollar of cost) is lognormal

$$\ln(Q_\ell / C_{i\ell}) \sim N\left(m_{s(\ell)}, s^2\right), \qquad s(\ell) \in \{\text{onshore}, \text{deep water}\}$$

  - Independent across firms and blocks
- Auctions for different blocks are independent

Bidding and Drilling Model



Drilling cost C_i (one-time, idiosyncratic)
Oil price $\{P_t\}$ (random walk, price taker)
Expected output $\{q_{it}\}$ (common to all bidders)

The Auction and the Post-Auction Drilling Decision

Stage 1. Entry. A firm knows Q_ℓ, its own cost $C_{i\ell}$, the bid limits $[r_{\min}, r_{\max}]$, the entry cost c_ℓ and the current price $\bar{P}_\ell$, and decides whether to enter.

Stage 2. Bidding. Each entrant submits a revenue share r. The highest share wins.

Stage 3. Drilling and Production. Each month until the lease period ends, the winner decides whether to drill. With revenue $(1-r)P_tQ_\ell$ and n months left ($V_0 = 0$),

$$V_n(P_t; r, Q_\ell, C_{i\ell}) = \max\Big\{ \underbrace{(1-r)P_tQ_\ell - C_{i\ell}}_{\text{drill now}},\ \underbrace{\delta\, E_t\left[V_{n-1}(P_{t+1}; r, Q_\ell, C_{i\ell})\right]}_{\text{wait a month}} \Big\}$$

- Oil price $\{P_t\}$ is a random walk with $E_t[P_{t+1}] = P_t$
- Price high relative to cost: drill now. Price too low: wait a month
- Drilling costs $C_{i\ell}$ and earns the value of all future output at once

$$E_t\Big[\sum_{\tau\geq 0}\delta^\tau q_\tau P_{t+\tau}\Big] = P_t\sum_{\tau\geq 0}\delta^\tau q_\tau = P_tQ_\ell$$

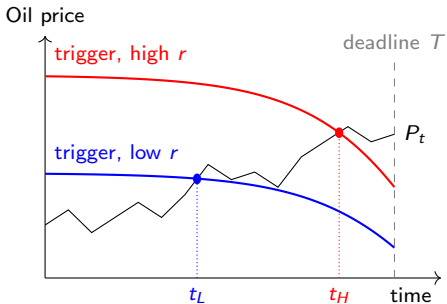
Payoffs if drilled in month t (zero otherwise): government $r\,P_tQ_\ell$, winner $(1-r)\,P_tQ_\ell - C_{i\ell}$

Proposition 1: Trigger Rule for Drilling

With n months left, the winner drills in month t iff

$$P_t \geq \underbrace{\frac{C}{Q_\ell}}_{\text{cost per barrel}} \times \underbrace{e^{y_n^*}}_{\text{option value}} \times \underbrace{\frac{1}{1-r}}_{\text{bid wedge}}$$

- Cost per barrel depends on who wins
  - A low-cost winner is more likely to drill
- Option value y^*_{T-t+1} falls as the deadline approaches
- A higher r raises the trigger, which delays or prevents drilling
  - A higher winning bid r lowers the winner's return on investment



A higher r delays drilling from t_L to t_H

Proposition 2: Optimal Bidding Anticipating Drilling

An entrant with cost C chooses its revenue share r at the contract-start price $\bar{P}_\ell$

$$\max_{r \in [r_{\min}, r_{\max}]} \underbrace{G_\ell(r)}_{\text{prob. of winning}} \quad \underbrace{V_T(\bar{P}_\ell; r, Q_\ell)}_{\text{value of the right to drill}}$$

- $G_\ell(r)$: probability that all rivals bid below r, with density $g_\ell(r)$

First-order condition

$$\underbrace{g_\ell(r)\, V_T(\bar{P}_\ell; r, Q_\ell)}_{\text{competition effect}} = \underbrace{G_\ell(r) \left(- \frac{\partial V_T(\bar{P}_\ell; r, Q_\ell)}{\partial r} \right)}_{\text{incentive effect}}$$

- Competition effect: a higher r wins more often
- Incentive effect: a higher r lowers revenue and delays drilling

Estimation: Maximum Likelihood of Bids and Drilling Years

Parameters θ to be estimated

- Mean and variance of firms' log cost efficiency $\ln(Q_\ell/C)$
- Variance of oil price shocks

Functions to be solved

- Equilibrium bid function $\rho_\ell(C_{i\ell})$
- Value function $V_n(P_t; r, Q_\ell, C_{i\ell})$

Set first, outside the likelihood

- Oil price volatility from 2010–2015 prices, discount factor δ
- Solve the value function by backward induction and compute the triggers y_n^*

Likelihood at each trial θ

1. Bids: cost distribution $\Rightarrow$ equilibrium bid function $\Rightarrow$ probability of observed bids
2. Drilling: winner's cost (from its bid) and realized prices $\Rightarrow$ probability of its drilling year
3. Choose θ to maximize the joint likelihood of bids and drilling years

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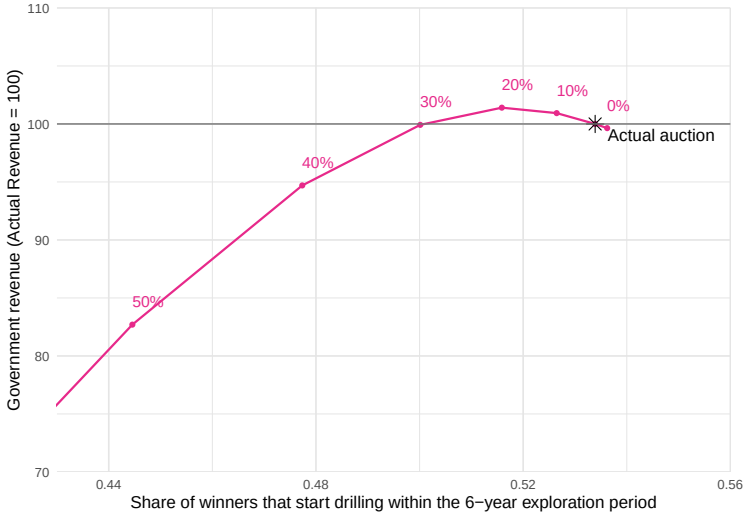
Counterfactuals: Alternative Auction Designs

Conclusion

How a Higher Minimum Bid Changes the Auction

- A minimum bid is a common policy tool to raise the winning bid
- But it also affects the winner's drilling decision
- Government revenue $= r\,P\,F(I)$
- Raising the minimum bid leads to $r \uparrow$ but $I \downarrow$ and $F(I) \downarrow$
- The government may lose revenue if $F(I) \downarrow$ outweighs $r \uparrow$
- Outcomes
  - Probability of drilling within the six-year lease period
  - Government revenue

Raising the Minimum Bid Can Reduce Government Revenue



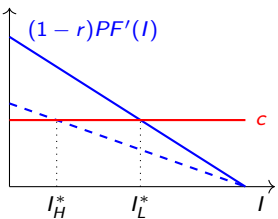
Auction with different minimum bids (%) and no ceiling

Toy Model: The Winner's Investment Incentives

Revenue-sharing ($100r$%)

$$\max_I \ (1-r)P\,F(I) - cI$$

$$(1-r)P\,F'(I^*) = c$$

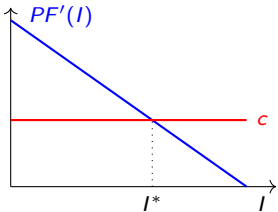


A higher r lowers investment

Fixed payment ($K)

$$\max_I \ P\,F(I) - cI - K$$

$$P\,F'(I^*) = c$$

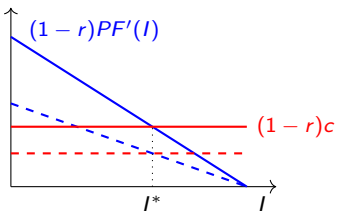


K does not enter the margin

Profit-sharing ($100r$%)

$$\max_I \ (1-r)\big(P\,F(I) - cI\big)$$

$$P\,F'(I^*) = c$$



r does not affect investment

Solid lines: low r. Dashed lines: high r.

Outcomes of Other Auction Designs

	Revenue share, no floor or ceiling	Bonus, first price	Profit share
Blocks awarded (%)	92.2	92.2	92.2
Winning bid	48% of revenue	$40.2 million	79% of profit
Winner drills within 6 years (%)	47.6	75.8	75.8
Winner's revenue (million $)	125.6	153.0	153.0
Winner's profit (million $)	40.3	72.9	21.2
Government revenue (million $)	59.3	37.0	88.7
Total surplus (million $)	99.6	110.0	110.0

Alternative Auctions

- A revenue share distorts the contractor's drilling incentive
- A fixed payment (bonus) or a profit share does not affect the drilling decision
- But both are hard to implement in emerging markets

Bonus auction

- The winner pays a fixed amount up front, so it bears all the reserve risk
- Insurance markets are thin in emerging markets

Profit-sharing auction

- The government must know the contractor's drilling cost
- Costs are hard to monitor, and the contractor has an incentive to overstate them

Conclusion

- In a revenue-sharing auction, the winning bid sets the winner's drilling incentive
  - A share r raises the trigger price by $1/(1-r)$, which delays or prevents drilling
- Reduced form: a higher winning bid delays exploration but raises government revenue
- Structural model: the auction screens for the lowest-cost firm
  - But spread-out shares and higher floors delay drilling
  - Raising the floor above 20% lowers government revenue
- Alternatives keep the drilling incentive (drilling within 6 years: 76% vs. 61%)
  - Bonus auction: less government revenue ($37m vs. $53m)
  - Profit-sharing auction: more government revenue ($89m vs. $53m)
  - But both are hard to implement in emerging markets

Auction design should account for the winner's incentives after the auction

References

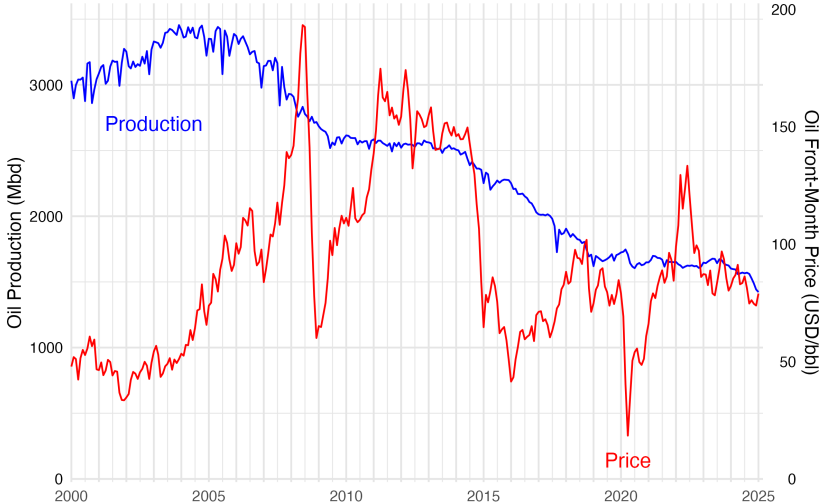
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Bhattacharya, V., Ordin, A., & Roberts, J. W. (2022). Bidding and drilling under uncertainty: An empirical analysis of contingent payment auctions. *Journal of Political Economy*, *130*(5), 1319–1363.

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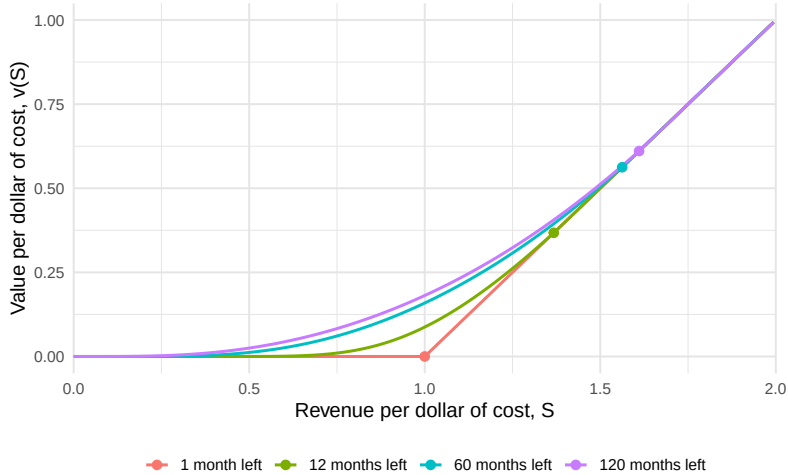
Production Does Not Track the Oil Price



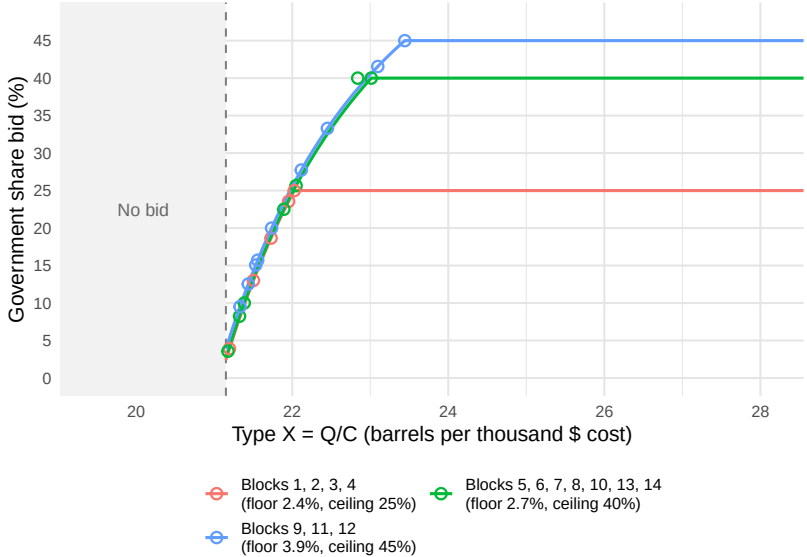
Model Fit: Predicted and Observed Outcomes

	Onshore		Deep water	
	Data	Model (SE)	Data	Model (SE)
Panel A. Entry and bidding, offered blocks				
Bids per offered block	2.67	2.67 (0.30)	1.33	1.35 (0.18)
Offered blocks with no bid	0.12	0.08 (0.05)	0.31	0.25 (0.07)
Bids at the ceiling	0.64	0.64 (0.06)	0.25	0.37 (0.07)
Mean interior share	0.16	0.21 (0.02)	0.12	0.13 (0.01)
SD of interior shares	0.10	0.11 (0.01)	0.06	0.06 (0.01)
Mean winning share	0.31	0.31 (0.01)	0.16	0.18 (0.01)
Winners at the ceiling	0.81	0.84 (0.07)	0.33	0.49 (0.08)
Panel B. Drilling, winners				
Winners that started drilling by 2024	0.57	0.77 (0.10)	0.11	0.16 (0.08)
Mean lag to the start of drilling (years)	2.33	1.89 (0.50)	5.00	3.55 (1.32)
Winners that started drilling by 2024, given actual winning share	0.57	0.78 (0.09)	0.11	0.16 (0.07)

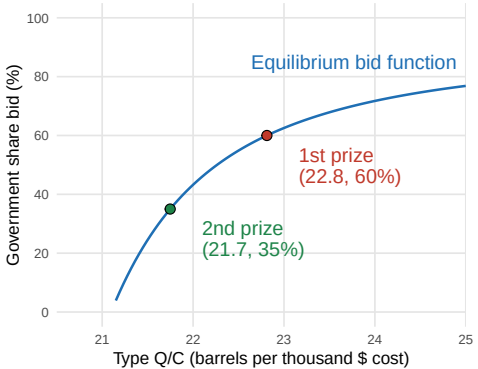
Value of the Right to Drill



Estimated Bid Function



The Second-Highest Bidder Can Bring More Revenue



	1st prize	2nd prize
Bid (revenue share %)	60	35
Q/C (bbl per \$1,000)	22.8	21.7
Drills within 6 years (%)	15.1	42.4
Bidder's exp. profit (million \$)	2.8	12.3
Gov's exp. rev. (million \$)	15.0	19.8
Total surplus (million \$)	17.8	32.0

- The incentive effect of r outweighs the effect of Q/C on drilling
- Ex-post inefficient allocation

Screening Effect of Revenue-Sharing Auctions

How much does the auction improve government revenue and drilling?

- The government wants to allocate a block to the lowest-cost company, but cannot observe drilling costs
- A revenue-sharing auction screens for the lowest-cost company
- Alternative to the auction: a bilateral revenue-share contract
  - Worst case: a randomly chosen company leases the block at a pre-set share r
  - First best: the lowest-cost company leases the block at a pre-set share r (infeasible)
- Outcomes: government revenue and drilling probability
- Costs of less drilling
  - Less production (the estimates capture only the extensive margin)
  - Less knowledge for future exploration of nearby fields

Screening Effect of Revenue-Sharing Auctions (%: Revenue Share)

